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Software Requirements Specification (SRS)

Private School Management System — Stage 1: Registration, Family & Fee Management

Version: 2.0 **Date:** August 6, 2026 **Status:** Draft for Review **Scope of this document:** Stage 1 only — organized module by module

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1. Introduction

1.1 Purpose

This document specifies the functional requirements for **Stage 1** of the Private School Management System, organized **module by module** for ease of development planning and estimation. It supersedes the Stage 1 section of the earlier consolidated SRS, and incorporates two changes agreed after that draft:

1. **Family-based registration and billing** — students are grouped under a Family unit, allowing combined household billing while retaining per-child detail.
2. **Student Promotion Module** — a new module allowing Admin to move selected students to their next grade/section at year-end.

1.2 Module Index

#	Module	Layer
M1	Academic Year & Term Setup	Foundation
M2	Grade & Section Management	Foundation
M3	User & Role Management	Foundation
M4	Fee Category & Structure Setup	Foundation
M5	Family & Guardian Management	Core
M6	Student Registration	Core
M7	Fee Counter (Payment Collection)	Core
M8	Event Management	Core
M9	Discounts & Scholarships	Supporting
M10	Recurring Fee Auto-Generation	Supporting
M11	Dues & Collection Dashboard	Supporting
M12	Receipts & Payment History	Supporting

#	Module	Layer
M13	Payment Reminders (Notifications)	Supporting
M14	Audit Log	Supporting
M15	Student Promotion	Core (new)

1.3 Definitions

Term	Meaning
Family	A household billing/registration unit that one or more Students and one or more Guardians belong to
Guardian	A parent/legal guardian; may belong to one Family and be linked individually to specific Students within it
Due Item	Any payable amount owed by a student (registration, monthly fee, event charge, etc.)
Enrollment	A record of which grade/section a student belonged to in a specific academic year
Promotion Batch	A draft set of students proposed for promotion, reviewed before being applied

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2. Foundation Modules

M1 — Academic Year & Term Setup

Purpose: Define the academic calendar that all other data is scoped to.

Key Entities: `AcademicYear (id, name, start_date, end_date, is_active)`, `Term (id, academic_year_id, name, start_date, end_date)`

ID	Requirement
FR-M1.1	Admin can create an Academic Year with start/end dates.
FR-M1.2	Admin can define Terms/semesters within an Academic Year.
FR-M1.3	Exactly one Academic Year is marked “active” at a time; all new registrations, fees, and enrollments default to it.
FR-M1.4	Only Admin can change the active Academic Year, and the system warns if any Stage 1 processes (e.g., an in-progress Promotion Batch) are incomplete before switching.

M2 — Grade & Section Management

Purpose: Define the structural units students are grouped into.

Key Entities: `Grade (id, name, sequence_order)`, `Section (id, grade_id, name, class_in_charge_id, capacity)`

ID	Requirement
FR-M2.1	Admin can create/edit Grades, each with a sequence order (used later by the Promotion module to determine "next grade").
FR-M2.2	Admin can create Sections within a Grade, optionally setting a capacity limit.
FR-M2.3	Admin can assign a Class-in-charge (Teacher) to a Section.
FR-M2.4	Grades and Sections persist across academic years; they are not recreated each year (a student's *enrollment* in a section is year-specific — see M15).

M3 — User & Role Management

Purpose: Manage staff accounts and access control.

ID	Requirement
FR-M3.1	Admin can create staff accounts (Admin, Accountant, Teacher) with role-based permissions.
FR-M3.2	Admin can add a Teacher and link the Subject(s) they are qualified to teach.
FR-M3.3	Admin can assign a Teacher to a specific Class/Section + Subject combination, restricted to subjects already linked to that teacher.
FR-M3.4	Access to Family, Student, and Fee data is restricted by role (e.g., Teachers cannot view financial details).

M4 — Fee Category & Structure Setup

Purpose: Configuration layer defining what can be charged and how much.

Key Entities: `FeeCategory (id, name, is_recurring)`, `FeeStructure (id, fee_category_id, grade_id, academic_year_id, amount, frequency)`

ID	Requirement
FR-M4.1	Admin can define Fee Categories (Registration, Tuition, Transport, Exam, etc.).

ID	Requirement
FR-M4.2	Admin can set a Fee Structure — amount per Grade, per Academic Year, with frequency (one-time / monthly / termly).
FR-M4.3	Fee Structures are versioned by Academic Year, so changing next year's tuition does not affect this year's already-generated dues.

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3. Core Modules

M5 — Family & Guardian Management

Purpose: Formalize the household unit so siblings can be registered together and billed as one household while remaining individually trackable.

Key Entities: `Family (id, family_code, address, home_contact_no, combined_billing_enabled)` `Guardian (id, family_id, name, relationship, contact_no, email, NIC, login_credentials)` `StudentGuardian (student_id, guardian_id)` — supports guardians linked to specific children within a family, not just the family as a whole

ID	Requirement
FR-M5.1	Admin/staff can create a new Family record, or search existing Families (by guardian name, phone, or NIC) to detect likely duplicates before creating a new one.
FR-M5.2	A Family can have one or more Guardians, and one or more Students.
FR-M5.3	Each Student within a Family can be linked to a subset of that Family's Guardians (supports blended families, e.g., a guardian relevant to only one child).
FR-M5.4	Family-level fields (address, home contact) are shared across all students in the family by default but can be overridden per student if needed.
FR-M5.5	Admin can toggle **Combined Family Billing** per family: ON shows one consolidated due balance across all children (itemized per child underneath); OFF keeps each child's dues fully separate. Default: ON.
FR-M5.6	A Family record is never deleted when a student graduates or leaves — it persists as long as at least one active or historical student remains linked, preserving payment history.
FR-M5.7	Guardian login credentials are tied to the Guardian record, and grant visibility only to the Students they are linked to within their Family (not automatically every child in the family, per FR-M5.3).

M6 — Student Registration

Purpose: Create and manage individual student profiles, linked to a Family.

Key Entities: `Student (id, name, DOB, gender, admission_no, photo, family_id, status)`

ID	Requirement
FR-M6.1	Staff can register a new student either as: (a) the first child of a **new Family**, or (b) an additional child added to an **existing Family** (searched via FR-M5.1).
FR-M6.2	When adding to an existing Family, shared fields (address, contact, guardians) are pre-filled and only student-specific fields (name, DOB, grade, photo, etc.) need to be entered.
FR-M6.3	On registration, the system auto-generates the applicable Registration Fee and any other mandatory one-time charges as Due Items for that student, per the active Fee Structure.
FR-M6.4	Student status remains **"Pending Registration"** until all mandatory Due Items generated at registration are fully paid — whether paid individually or as part of a combined family payment.
FR-M6.5	Only students with status **"Active"** are eligible for attendance marking and appear in class rosters.
FR-M6.6	Staff can edit student profiles, including changing section (mid-year transfer) or grade, with a change history retained.
FR-M6.7	Staff can mark a student inactive/withdrawn/graduated at any time (graduation is also handled in bulk via M15).

M7 — Fee Counter (Payment Collection)

Purpose: Day-to-day collection screen for staff, now family-aware.

ID	Requirement
FR-M7.1	Staff can search by Family or by individual Student to view outstanding Due Items.
FR-M7.2	For families with Combined Billing ON, staff see one total balance with a breakdown showing each child's contribution to that total.

ID	Requirement
FR-M7.3	Staff can record a payment (cash, bank, or online) against the family's combined balance or against an individual child's dues.
FR-M7.4	When a combined family payment does not fully settle the total balance, the system applies a configurable **allocation strategy** to distribute the payment across children's Due Items: (a) **Evenly split** across all children with outstanding dues, (b) **Oldest-child-first** (by admission order or grade), (c) **Manual allocation** — staff/parent selects which child's dues to prioritize.Default strategy is configurable by Admin at the school level; manual allocation is always available as an override at the counter.
FR-M7.5	Every payment generates a receipt itemizing exactly which Due Items (and which child, if combined) were settled.
FR-M7.6	Partial payments update each affected Due Item's balance individually, even when made as one combined transaction.

M8 — Event Management

Purpose: Create school events with associated charges.

ID	Requirement
FR-M8.1	Admin can create an Event with name, date, description, and applicable grade(s).
FR-M8.2	Admin can set a charge per grade — uniform or grade-specific amounts.
FR-M8.3	Admin can mark an event as opt-in or mandatory.
FR-M8.4	On confirmation (or automatically for mandatory events), the system generates a Due Item per applicable student, which then participates in Family combined billing exactly like any other fee (per M5/M7).
FR-M8.5	Admin can view participation and payment status per event, at both student and family level.

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4. Supporting Modules

M9 — Discounts & Scholarships

Key Entities: `Discount (id, student_id, type, value, applies_to_fee_category_id)`

ID	Requirement
FR-M9.1	Admin can apply a discount (sibling, scholarship, staff, other) to a specific student's fee category.
FR-M9.2	A **sibling discount** can optionally be auto-suggested when a Family has 2+ active students, based on a configurable rule (e.g., 10% off tuition from the 2nd child onward).
FR-M9.3	Discounts are reflected at Due Item generation, before any combined billing or payment allocation logic runs.

M10 — Recurring Fee Auto-Generation

ID	Requirement
FR-M10.1	The system automatically generates recurring Due Items (e.g., monthly tuition) per the active Fee Structure, on a scheduled date each cycle.
FR-M10.2	Auto-generated Due Items respect each student's applicable discounts and opt-in fee categories (e.g., only students enrolled in transport get a transport charge).
FR-M10.3	Newly generated Due Items feed into Family combined billing automatically — no manual step required.

M11 — Dues & Collection Dashboard

ID	Requirement
FR-M11.1	Admin/Accountant can view total due, total collected, and outstanding balance — filterable by grade, section, date range, or fee category.
FR-M11.2	Dashboard supports a **Family-level view**: which families have outstanding balances, useful since one guardian contact often covers multiple children's dues.

M12 — Receipts & Payment History

ID	Requirement
FR-M12.1	Every payment produces a stored, reprintable receipt.
FR-M12.2	Full payment history is viewable both per student and per family.

ID	Requirement
FR-M12.3	A combined family payment's receipt clearly itemizes the per-child breakdown, even though it was one transaction.

M13 — Payment Reminders (Notifications)

ID	Requirement
FR-M13.1	Automated reminders are sent to Guardians when a Due Item is upcoming or overdue.
FR-M13.2	For families with Combined Billing ON, reminders are consolidated into a single message per family rather than one per child, to avoid spamming guardians.

M14 — Audit Log

ID	Requirement
FR-M14.1	The system logs who registered a student, who created/edited a family, who applied a discount, and who recorded or modified a payment, each with a timestamp.
FR-M14.2	Changes to payment allocation strategy (FR-M7.4) on a specific transaction are logged, since this affects which child's balance is considered settled.

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5. New Module: Student Promotion

M15 — Student Promotion

Purpose: At the end of an academic year, allow Admin to move selected students in bulk from their current grade/section to the next grade/section for the upcoming academic year — without altering any historical record (attendance, marks, fees) from the year that just ended.

Key Design Decision: A student's grade/section must **not** be a single fixed field that gets overwritten each year. Instead, introduce an `Enrollment` entity so each year's grade/section assignment is its own historical record:

Key Entities: `Enrollment (id, student_id, academic_year_id, grade_id, section_id, status)` — status: *promoted / retained / transferred / graduated* `PromotionBatch (id, source_academic_year_id, target_academic_year_id, created_by, status)` — status: *draft / confirmed* `PromotionBatchItem (id, promotion_batch_id, student_id, source_section_id, target_grade_id, target_section_id, action)` — action: *promote / retain / graduate / exclude*

ID	Requirement
FR-M15.1	Admin selects a source Academic Year and one or more Sections to begin a Promotion Batch.
FR-M15.2	The system lists all Active students in the selected section(s), with a **default target grade/section** pre-filled as “next sequential grade” (based on Grade `sequence_order` from M2), same section letter where possible.
FR-M15.3	Admin can review the list and, per student: keep the default promotion, change the target section (e.g., redistribute into a different section), mark as **Retained** (stays in the same grade next year), mark as **Excluded** (not moved — e.g., pending transfer decision), or mark as **Graduated** (for the final grade level).
FR-M15.4	The Promotion Batch is created in **draft status** — nothing is applied to student records until Admin explicitly confirms. Draft batches can be edited, re-reviewed, or discarded.
FR-M15.5	On confirmation, the system creates a new `Enrollment` record for each promoted/retained student for the target Academic Year, **without modifying** their Enrollment record from the source year.
FR-M15.6	For students marked **Graduated**, the system sets their overall student status to “Graduated” and does not create a new Enrollment; the student’s full historical record (fees, attendance, marks) remains accessible via the Family and Student profile.
FR-M15.7	Promotion does **not** automatically generate next year’s fee Due Items — those continue to be generated normally by M10 once the target Academic Year becomes active, avoiding double-charging or premature billing.
FR-M15.8	A confirmed Promotion Batch can be exported (CSV/PDF) as a record of which students moved where — useful for office records and any promotion/leaving certificates.
FR-M15.9	Admin can view a student’s full **Enrollment History** (which grade/section they were in each academic year) from their profile at any time.

ID	Requirement
FR-M15.10	Class-in-charge assignments (M2) are re-evaluated per target section for the new year; promotion does not assume the same teacher remains in charge of “Grade 6A” if that teacher was actually attached to last year’s “Grade 5A.”

6. Cross-Module Data Model Summary

Entity	Belongs To Module	Key Relationships
AcademicYear, Term	M1	Referenced by nearly every other entity for scoping
Grade, Section	M2	Referenced by Enrollment (M15), TeacherAssignment (M3)
Family	M5	Has many Students, has many Guardians
Guardian	M5	Belongs to Family; linked to specific Students via StudentGuardian
Student	M6	Belongs to Family; has many Enrollments (M15); has many Due Items
FeeCategory, FeeStructure	M4	Drives Due Item generation (M6, M8, M10)
DueItem	M6/M8/M10	Belongs to Student; aggregated at Family level for billing (M7)
Payment	M7	Settles one or more Due Items; may span multiple students in a Family
Event, EventFeeAssignment	M8	Generates Due Items per applicable student
Discount	M9	Applied at Due Item generation
Enrollment	M15	One per Student per Academic Year — the historical record of grade/section
PromotionBatch, PromotionBatchItem	M15	Draft-then-confirm workflow producing new Enrollment records

7. Non-Functional Notes Relevant to This Stage

- **Data integrity:** Confirming a Promotion Batch must be a single atomic operation — either all selected students’ Enrollments are created, or none are, to avoid a half-promoted

cohort.

- **Reversibility:** A confirmed Promotion Batch should be reversible by Admin within a safety window (e.g., before any new-year attendance or fees are recorded against the new Enrollments), rather than requiring manual data correction.
- **Family privacy:** Guardians linked to only one child in a multi-child Family (per FR-M5.3) must never see due items, attendance, or academic data for children they are not linked to, even under combined billing.
- **Auditability:** All Family creation/merge actions and all Promotion Batch confirmations are logged per M14.

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8. Open Items for Confirmation

1. Default payment **allocation strategy** for partial combined-family payments (FR-M7.4) — evenly split, oldest-first, or manual-only by default.
2. Default **sibling discount** percentage and rule (FR-M9.2), if this should be automatic rather than manually applied each time.
3. **Promotion safety window** — how long after confirming a Promotion Batch should reversal remain possible (FR-M15's non-functional note).
4. Whether **duplicate family detection** (FR-M5.1) should block registration outright on a likely match, or just warn and let staff proceed.

End of Document — Version 2.0